

Data Warehouse and Data Mining

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Unit-1

Unit-2

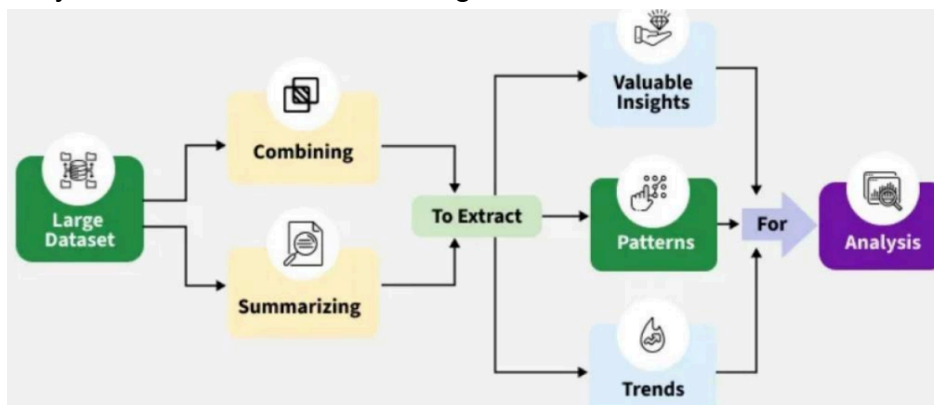
Unit-3

Unit-4

Unit-5

Aggregation**

- combining + summarizing large amounts of data into smaller meaningful form
Why need? Raw data are too big, detailed and difficult to understand.



Examples -> Daily Sales into weekly sales, marks of student -> average marks.

Data Aggregator

- Tool that collect data, combines, summarizes and give useful patterns.

Working of Data Aggregator?

- Collect data from different source
- combines data
- summarize data - use functions

Why Data Generalization needed?

- summarize, find patterns, remove unnecessary details, decision making.

Analytical Characterization

- descriptive summarize of data, using functions like (mean, median, mode etc)
Historical Information?

- old data stored in data warehouse, that help use to understand trends, patterns and change over time.
- Why? -> sales, customer behavior, profit, trends analysis change over time, decision making, forecasting

Year	Sales
2020	5 lakh
2021	7 lakh
2022	9 lakh
2023	10 lakh
2024	8 lakh

From historical data, the shop can see:

- Sales were growing till 2023
- Sales dropped in 2024
- Festival months perform better
- Summer sales are low

These insights are **not possible** with only current data.

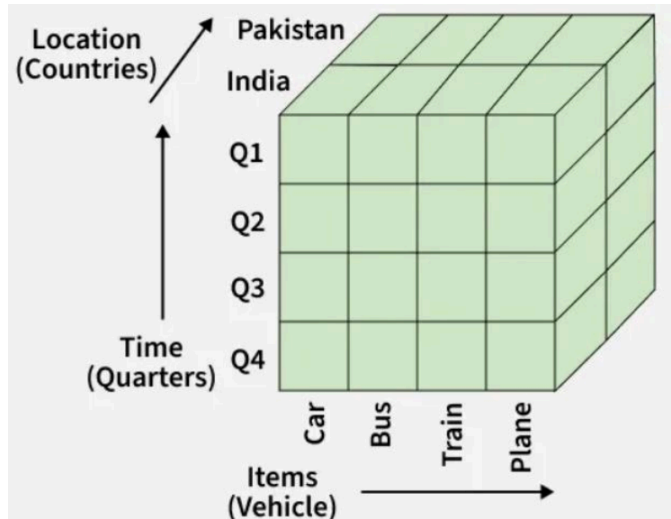
Query Facility

- tool that helps customer to get information for their query from data warehouse.
- Types?
 - 1. Simple Queries (list all products)
 - 2. Complex Queries (last 3 months sales)
 - 3. OLAP Queries (deep analysis)

OLAP

- online analytical processing
- technology used to analyse large amount of data in warehouse
- view from different dimensions for better decisions.

- **OLAP Cube** -> stores data in multiple dimensions



Types of OLAP operations -

- 1. Roll-up (summarization): Day -> Month -> Year
- 2. Drill-down (detailing): Year -> Month -> Day
- 3. Slice (single values of one dimension): cube to 2D table
- 4. Dice (subset of dimension):
- 5. Pivot Rotate (rotate data to view from different angle)

- **OLAP Servers** -> system that process OLAP queries

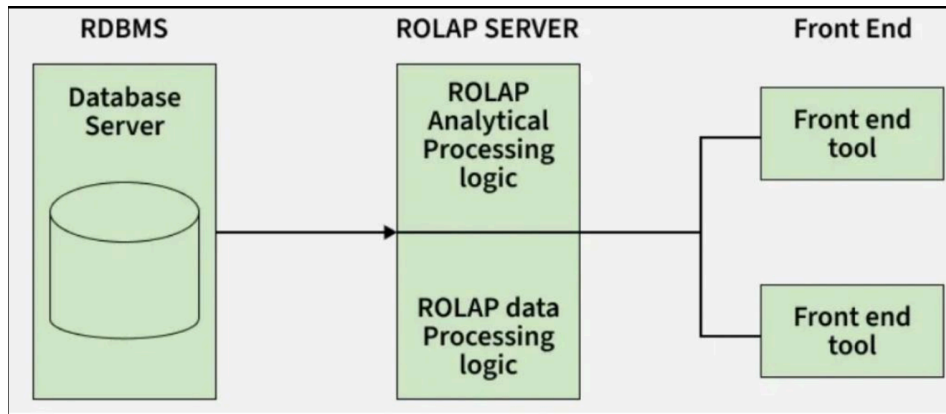
- **Types of OLAP Servers**

- 1. ROLAP (relational)
- 2. MOLAP (multidimensional)
- 3. HOLAP (hybrid)

ROLAP

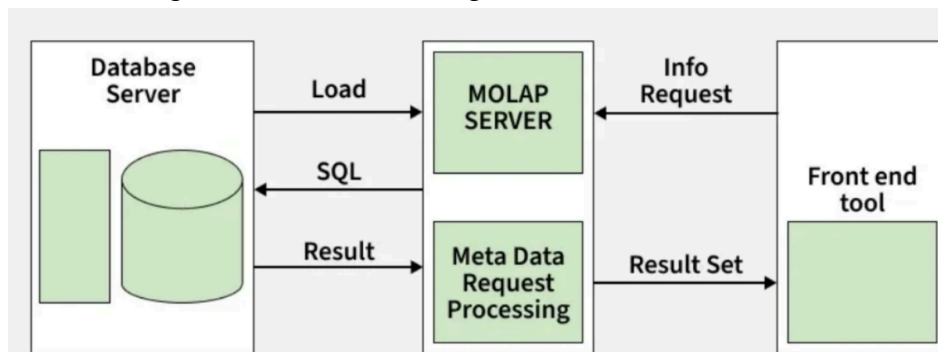
- store data in relations database + SQL for analysis
- Working: Data store in tables, ROLAP server convert OLAP query into SQL, Runs on DB, response to user
- Advantage: handle large data + RDBMS + SQL + no cube storage

- Disadvantage: slower than MOLAP + depends on SQL



MOLAP

- store data in multi-dimensional cube
- precompute summary data
- Working? -> data load from DB, MOLAP server create cubes, user query cubes and instant result.
- Advantage -> Fastest + complex queries + instant result
- Disadvantage -> can't handle large data + cube creation take time and memory



HOLAP

- ROLAP + MOLAP
- Working? -> detail data query from SQL + summary result from cube
- Advantage -> handle large data + better than ROLAP + flexible than MOLAP
- Disadvantage -> complex + costlier

Difference between these three

Feature	ROLAP	MOLAP	HOLAP
Storage	Relational tables	Multidimensional cubes	Both
Speed	Medium	Fastest	High
Handles Big Data?	Yes	No	Yes
Query language	SQL	Cube queries	SQL + cubes
Best for	Large data warehouses	Fast analysis	Mixed workloads

E.F Codd's 12 OLAP Rules

- system must follow these rules in order to call OLAP System
-
- 1. Multidimensional view (time, product, location)
-
- 2. Transparency (no overhead)
-
- 3. Accessibility (one interface)
-
- 4. Consistent report performance
-
- 5. Client-server architecture
-
- 6. Generic Dimensionality (all dimensions should have similarity)
-
- 7. Sparse handling (handling of empty cells in cube)
-
- 8. cross-dimension operations
-
- 9. multi user support
-
- 10. Intuitive data manipulation
-
- 11. flexible reporting
-
- 12. dimensions and aggregation levels

Data Mining Interface

- screen, dashboard, tool that allow user to interact with data mining system
- bridge between user and data mining system

Role of Data mining interface?

1. select data
2. data mining tasks
3. runs algorithms
4. results visually
5. drill down and drill up
6. save and export
7. user friendly usage

Data Backup and Recovery

- protect from system failures, hardware issue, human errors etc
- **Data Backup** -> copy of data and store it, (full, incremental, schedule, off-site, replication) backups
- **Data Recovery** -> restoring lost or damaged data from backup. load backup files from warehouse, missing data rebuild using logs, warehouse run again after failure
- **Tuning** -> improving performance of data warehouse so that queries run faster
- **Testing** -> checking whether data warehouse is working correctly, complete, accurately

How Tuning is done?

1. Indexing -> speed up
2. Partitioning -> big tables into small tables
3. Query Optimization
4. aggregated tables
5. hardware tuning
6. Caching
7. Load Balancing

Types of Data warehouse testing?

1. Data accuracy testing
2. data completion testing
3. ETL testing
4. Performance testing
5. Integration testing
6. Security testing
7. User acceptance testing

Applications of Data warehousing?

- Business, marketing, finance, retail, e-commerce, healthcare, government, telecommunication, education

Security in Data Warehouse

- protecting sensitive data from unauthorized access
- authentication, authorization, data encryption, access control, audit logs, firewall and network security

Web Mining

- finding useful patterns from web data (webpages, logs etc)
- why? -> improvement, better search results, target ads

Types of Web Mining

1. web content - from web pages
2. web structure - study links between webpages
3. web usage - user behavior
4. spatial - data related to location or geographical
5. temporal - data that changes over time

Difference between Spatial & Temporal Mining

Feature	Spatial Mining	Temporal Mining
Meaning	Mining data related to location or geography	Mining data related to time
Focus	"Where" things happen	"When" things happen
Type of Data	Maps, GPS, satellite data, coordinates	Time-series data (daily, monthly, yearly)
Examples	Crime hotspots in a city, disease spread in areas	Sales trend over years, weather change over time
Goal	Find spatial patterns based on location	Find time-based patterns, trends, sequences
Applications	City planning, agriculture, disaster management	Forecasting, finance, stock analysis
Output	Spatial clusters, geographic patterns	Time trends, sequences, periodic behavior
Important Question	"Where does it occur?"	"When does it occur?"